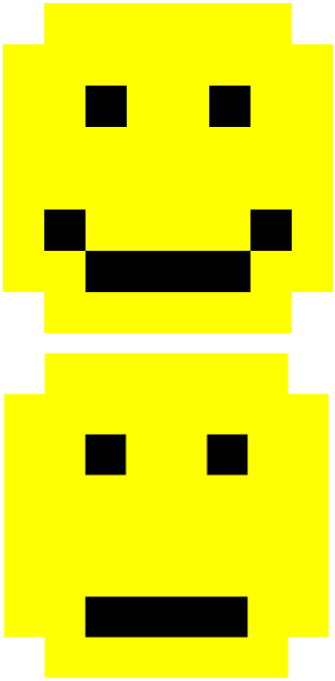
ASTRO PI[1] MISSION ZERO
Scratch Sensor Simulator

Send a personalised image to the astronauts on the Space Station!



Last week you created two 8x8 pixel-art works. Pixels are picture elements.

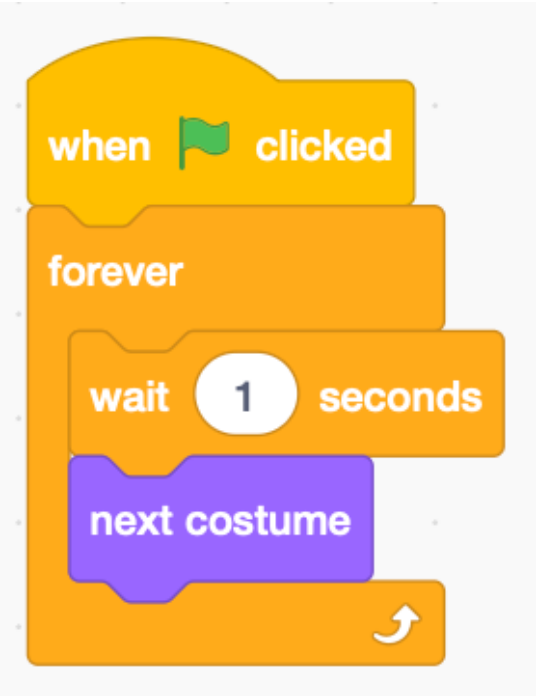
1) Open **pixilart.com** – login as **CodeClub67**
2) For each of your Pixilart images:
   a) Click on the smiley face at top-right
   b) Open **My Gallery**
   c) Select your image – you can **edit** it if you like.
   d) In the panel on the right scroll down to **Details**
   e) Click **Download Original** to download a .png (ping).
   f) Save it in your **downloads** folder.

Your mission is to animate your images in Scratch.

3) Register/login at https://scratch.mit.edu
4) Create a new Scratch project
5) **Upload** a new sprite from the first image in downloads.
6) It's very small – only 8x8. Adjust the **size** to 1000%.
7) On the **costumes** tab upload the second image.

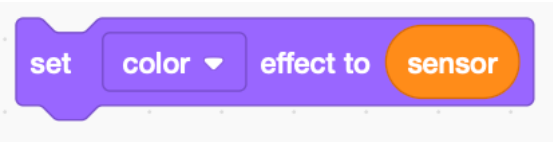
[1] astro-pi.org

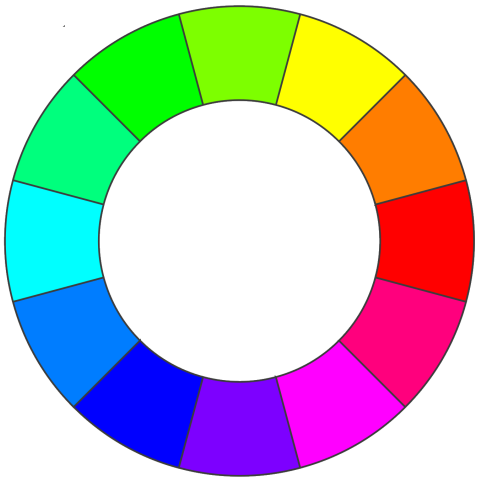
8) Add the code (right) to animate the sprite by switching between the two costumes every second.



Simulate a space station sensor.

9) **Make a Variable** called **sensor** (for all sprites).
10) Make sure the variable is **ticked** so it's shown on-screen.
11) Right-click the on-screen variable and make it a **slider**.
12) Right-click the on-screen variable to **change slider range.** Set the **maximum value** to 200 – **OK**
13) Add this block into the loop to set the colour effect according to the sensor data.





*Colours can be placed on a colour wheel. The Scratch **colour effect** shifts any colour clockwise around the wheel.*

The wheel is 200 units around, so a value of 100 will shift any colour to the opposite side.

Try a value of 100, does it shift the colour to the opposite side of the colour wheel?

Try a value of 200 – what happens? A sensor value of 200 takes the colour all the way round back to where it started.

*Remember to **Save** your code with a good name.*